

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2023-2024
FULL MARKS: 75

CSE 4849: Human Computer Interaction

Programmable calculators are not allowed. Do not write anything on the question paper.
 Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Suppose you are developing a mental health therapy chatbot designed to provide conversational support to users experiencing stress, anxiety, or depression. The chatbot engages in dialogue by detecting emotional cues from user messages, maintaining context across multiple sessions, and referring back to past interactions. It draws on a knowledge base of therapeutic techniques, including cognitive behavioral therapy, mindfulness, and active listening strategies. It evaluates the users progress, adjusts its responses, and recommends coping strategies based on patterns it has observed and user feedback. Answer the following:
 - i. Map the chatbot's actions to the cognitive processes of perception, attention, memory, and knowledge. Explain how these processes are simulated to enable meaningful and empathetic mental health conversations. 7
(CO1)
(PO1)
 - ii. The chatbot is designed to simulate reflective cognition, providing personalized and adaptive support. How can reflective thinking be built into the system? 8
(CO1)
(PO1)
- b) Consider the Cognitive Architecture for Language Agents (CoALA) in Figure 1 proposed by Summers et. al. from Princeton University. Identify and design each CoALA module in the system by creating an interaction flowchart showing how the chatbot would process, remember, adapt, and act during a single therapy session. 10
(CO2)
(PO3)

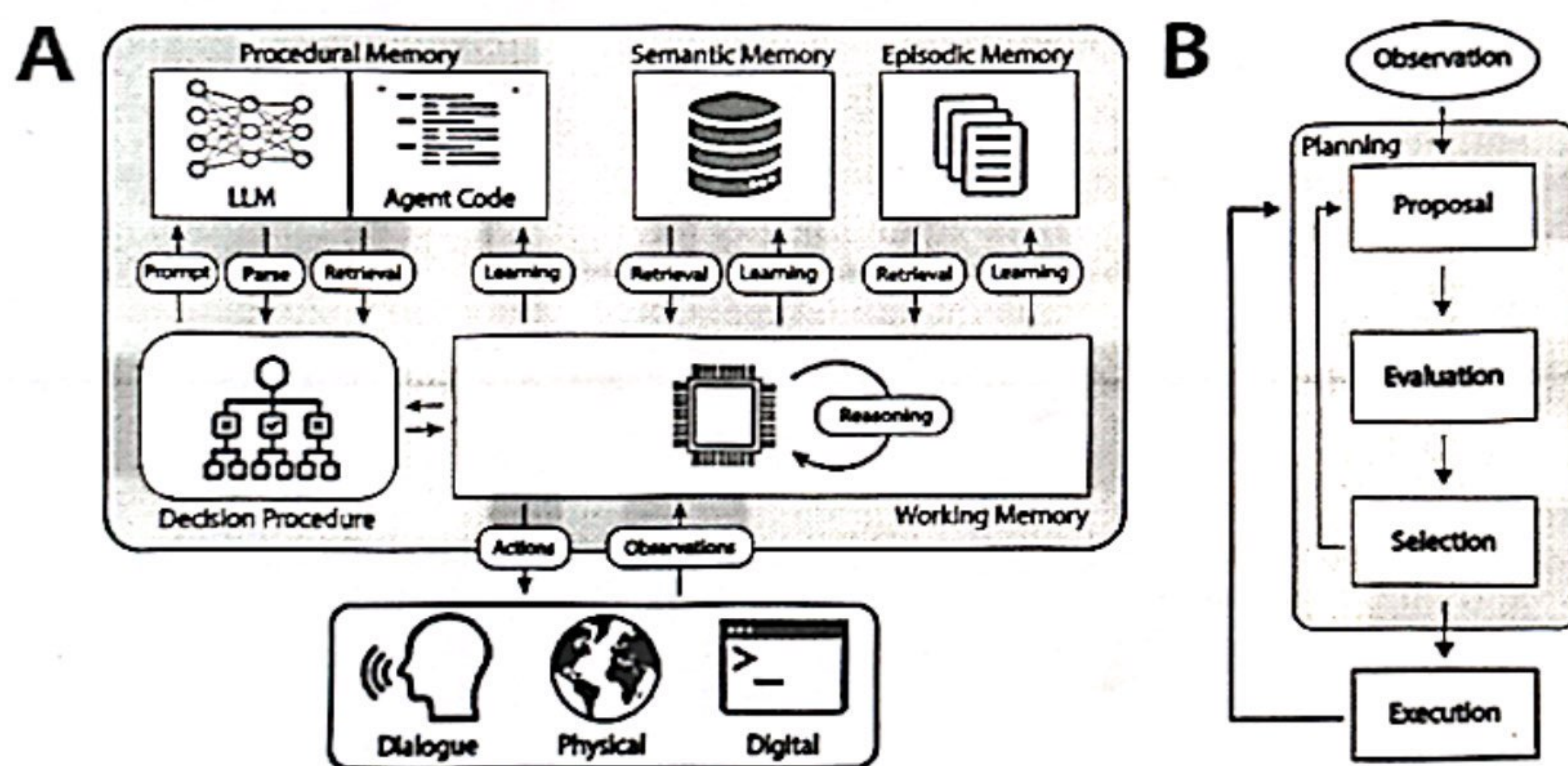


Figure 1: Cognitive architectures for language agents (CoALA). **A:** CoALA defines a set of interacting modules and processes. **B:** Temporally, the agents decision procedure executes a decision cycle in a loop with the external environment.

2. a) In human visual perception, cone cells handle central vision, which is responsible for recognizing fine details, while rod cells manage peripheral vision, detecting motion and broader visual cues. Our ability to distinguish fine detail decreases as objects move away from our central vision. This principle has design implications that help ensure the app is both effective and safe. Consider the following list of tasks while designing a map navigation app for the car drivers: 10
(CO2)
(PO2)

- Reading Estimated Time of Arrival (ETA).
- Noticing a change in the color of the route line (e.g., red for heavy traffic)
- Following Turn-by-Turn text instructions
- Observing incoming call or message notification banners during navigation
- Tapping the screen to search for a new destination

For each task, consider the design implications and decide whether it should be designed for central vision (cone cells) or peripheral vision (rod cells). Provide a one-line justification for your choice.

- b) In a graphical user interface, a pointing task involves four targets labeled as A, B, C, and D, placed at varying distances from the starting point. The distances (D) from the starting point to the center of each target are as follows: $D_A = 180$ pixels, $D_B = 160$ pixels, $D_C = 180$ pixels, and $D_D = 140$ pixels. The widths (W) of the target areas are: $W_A = 20$ pixels, $W_B = 18$ pixels, $W_C = 80$ pixels, and $W_D = 15$ pixels. Constants for movement time (MT) calculations are, $a = 90$ ms, $b = 40$ ms. Calculate human throughput using Fitts' law for each target and determine which target provides the highest throughput. 15
(CO2)
(PO1)
3. a) Bangla word-level sign language, 'BdSLW60' dataset, contains 60 isolated Bangla sign language words with 9307 annotated gesture trials from 18 signers. Each sign is captured as a video and includes variations in signer style, hand shape, and motion. 13
(CO2)
(PO2)
- Gestures are described using landmark coordinates (hand, face, body) and temporal parameters like speed, duration, and direction.
- To represent the meaning and structure of each gestural word, we can use techniques such as semantic networks, frame-based, script-based, or rule-based knowledge representation. For example, the sign for "Abba (Father)" involves raising the right hand near the forehead, moving it forward slightly with fingers open. The gesture lasts about 1.2 seconds and starts from the right shoulder, with the palm facing inward.
- Which knowledge representation technique (e.g., semantic network, frame-based, script-based, or rule-based) is most appropriate for modeling the signs? Justify your answer with reasoning and briefly describe how the gesture would be represented. [Hint: You can use a JSON-like structure to represent the gesture and its key parameters]
- b) You are hired as an interaction designer to evaluate user interactions in a newly developed fitness tracking app. This app allows users to record workouts, monitor daily steps, and manage personal goals using touch gestures on smartphones. Users input commands through the app, and the system provides feedback on their activities. Based on the Abowd and Beale interaction framework, identify and explain the interaction problems (articulation, performance, presentation, and observation) that may exist (one or more) for the following tasks within 1-2 line justification for each problem you found. 12
(CO3)
(PO1)
- Users try to log their workout sessions by selecting exercise types and durations, but some users find it difficult to select precise exercise categories, leading to inaccurate workout logs. There are also occasional delays when saving session data.
 - Users navigate to view their weekly step count and goal progress. Some users report that the graphs and statistics are confusing and hard to interpret, making it difficult to understand if they are meeting their fitness goals.
 - Users attempt to set reminders for daily hydration but miss seeing the confirmation alert after enabling reminders, causing uncertainty about whether the reminders were successfully set.
 - Users report that when starting a live workout tracking session, the app occasionally freezes or crashes, forcing them to restart the app and lose their current session data.

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MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2023-2024
FULL MARKS: 75

CSE 4803: Graph Theory

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Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Figure 1 shows a weighted graph G .

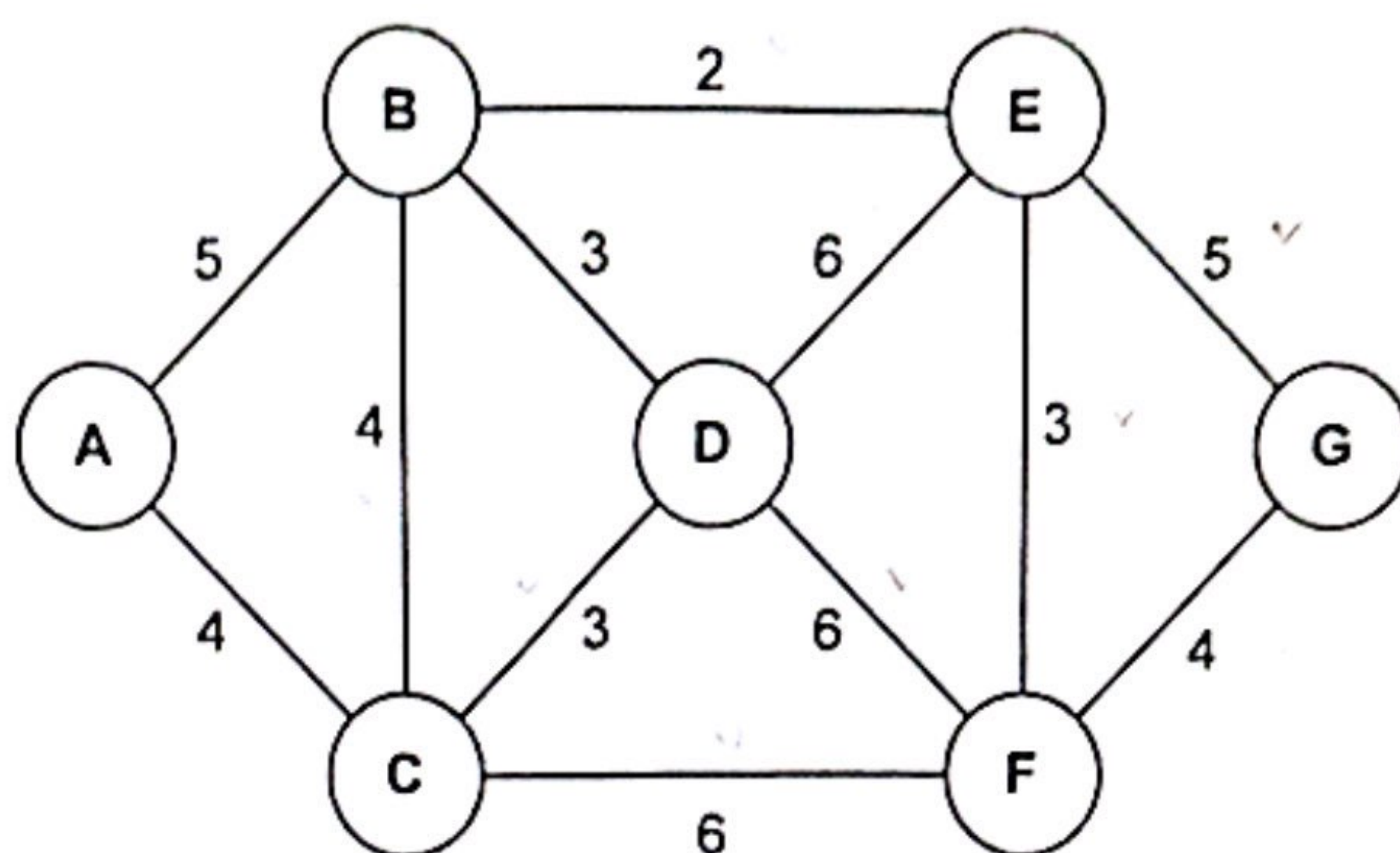


Figure 1: A sample weighted graph containing 7 vertices for Question 1.a.

- | | |
|---|----------------------------------|
| <p>i. Find the minimum spanning tree of G using both Prim's algorithm and Kruskal's algorithm. Show the detailed steps of the process you followed. Assume vertex A as the initial vertex for running Prim's algorithm.</p> | <p>6 + 6
(CO1)
(PO1)</p> |
| <p>ii. What condition needs to be fulfilled so that both Prim's algorithm and Kruskal's algorithm always produce the same minimum spanning tree?</p> | <p>3
(CO1)
(PO1)</p> |
| <p>b) Let T_1 and T_2 be two different spanning trees of a graph G. Let e be an edge in T_1 that does not exist in T_2, and f be an edge in T_2 that does not exist in T_1. Prove that both $T_1 - e + f$ and $T_2 - f + e$ are spanning trees of G.</p> | <p>10
(CO1)
(PO1)</p> |
| <p>2. a) Suppose a graph G has 6 vertices, $x_1, x_2, \dots, x_6$. You are given the following information regarding their degrees: $\deg(x_1) = 5$, $\deg(x_2) = 4$, $\deg(x_3) = 3$, $\deg(x_4) = 2$, $\deg(x_5) = a$, and $\deg(x_6) = b$, where a and b are unknown values. Among all the values of (a, b) for which such a graph exists, determine the minimum possible value of $a + b$. Also, draw one such graph.</p> | <p>8 + 3
(CO1)
(PO1)</p> |
| <p>b) Write short notes on the following:</p> <ul style="list-style-type: none"> i. Vertex connectivity ii. Edge connectivity iii. k-regular graph | <p>3 × 3
(CO1)
(PO1)</p> |
| <p>c) Find the smallest 3-regular graph with a vertex connectivity of 1.</p> | <p>5
(CO1)
(PO1)</p> |

3. a) Suppose a round-robin tournament is being played among 6 players, named P_1 through P_6 . Every player needs to play against every other player. At a given time slot, multiple games can be conducted among different players but the same player cannot participate in more than one game in the same time slot.

i. Formulate this scenario as a graph theoretic problem.

6
(CO2)
(PO2)

ii. Recommend a solution to find the minimum number of time slots we need to schedule all the games in the tournament.

6
(CO3)
(PO3)

- b) You are given three vessels A , B , and C of capacities 8, 5, and 3 gallons, respectively. Initially, A is filled, while B and C are empty. Your task is to divide the liquid in A into two equal quantities by pouring the liquid from one vessel to another. It should be noted that when pouring the liquid, you cannot stop pouring until either the pouring vessel is empty or the vessel that is being poured into becomes full. You cannot use any additional tools. Your objective here is to find the sequence of actions that would take you from the initial state ($A = 8, B = 0, C = 0$) to the goal state ($A = 4, B = 4, C = 0$).

i. Formulate the problem as a graph by identifying the vertices and edges.

6
(CO2)
(PO2)

ii. Recommend one possible solution to this problem.

7
(CO3)
(PO3)

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SUMMER SEMESTER, 2023-2024
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CSE 4801: Compiler Design

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Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

-
1. a) Discuss the role of a lexical analyzer during compilation process along with a block diagram. 10
(CO1)
(PO1)
 - b) Describe the phases of a compiler. 10
(CO1)
(PO1)
 - c) Differentiate tokens, patterns, and lexemes with examples. 5
(CO1)
(PO1)
 2. a) A grammar is shown in Equation (1):

$$L \rightarrow L, S$$

$$L \rightarrow S$$

$$S \rightarrow (L)$$

$$S \rightarrow a$$

(1)

 - i. Construct a predictive parse table for the the grammar. 14
(CO2)
(PO2)
 - ii. Find the canonical collection of sets of LR(0) items for the grammar. 6
(CO2)
(PO2)
 - b) Define left recursion. When and why elimination of left recursion is required? 5
(CO2)
(PO1)
 3. a) Explain the roles of symbol table manager and error handler during compilation. 8
(CO2)
(PO1)
 - b) Write short notes on handle and handle pruning . 5
(CO2)
(PO1)
 - c) A grammar is presented below in Equation (2) and the respective SLR parse table is shown in Table 1. Now, step by step show the SLR parsing process for the input string (id+id)*id, where id is a terminal symbol. 12
(CO2)
(PO2)

$$\begin{aligned}
 E &\rightarrow E + T \\
 E &\rightarrow T \\
 T &\rightarrow T * F \\
 T &\rightarrow F \\
 F &\rightarrow (E) \\
 F &\rightarrow id
 \end{aligned}
 \tag{2}$$

Table 1: An SLR parse table for Question 3.c.

State	id	+	*	(	)	\$	E	T	F
0	s5			s4			1	2	3
1		s6				acc			
2		r2	s7		r2	r2			
3		r4	r4		r4	r4			
4	s5			s4			8	2	3
5		r6	r6		r6	r6			
6	s5			s4				9	3
7	s5			s4					10
8		s6			s11				
9		r1	s7		r1	r1			
10		r3	r3		r3	r3			
11		r5	r5		r5	r5			

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SUMMER SEMESTER, 2023-2024
FULL MARKS: 50

CSE 4809: Algorithm Engineering

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Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

-
- | | | |
|----|---|---|
| 1. | <p>a) Answer the following questions in brief:</p> <ul style="list-style-type: none"> i. Why do we look for asymptotic bound of any algorithm? ii. Mention the characteristic(s) of algorithms that have a complexity of $\lg(n)$. iii. How does divide and conquer technique help in the merge sort algorithm? iv. Why does quick sort algorithm just have division (i.e. partition) but does not have any merging? <p>b) Write down the rationale of master method for its three cases for determining the complexity of recursive algorithms.</p> <p>c) Given a regular recursion:</p> $T(n) = aT(n/b) + f(n)$ <p>If $f(n) = O(n^{\log_b a - \epsilon})$, prove that $T(n) = \theta(n^{\log_b a})$.</p> | <p>2×4
(CO2)
(PO2)</p> <p>5
(CO1)
(PO1)</p> <p>5
(CO1)
(PO1)</p> |
| 2. | <p>a) Answer the following questions in a single sentence:</p> <ul style="list-style-type: none"> i. How does dynamic programming save computation of a combinatorial optimization problem? ii. Can dynamic programming be used in a path finding problem where the problem is to find the list of paths within a ratio of the optimal paths? iii. Does Traveling Salesman Problem (TSP) have a dynamic programming solution? iv. What is optimal substructure equation? v. Why longest simple path finding problem does not have an optimal substructure property? <p>b) Write down the optimal substructure equation for Longest Common Subsequence (LCS) after briefly describing the optimization problem the algorithm attempts to solve.</p> <p>c) Comment on the complexity of Dynamic Time Warping (DTW) algorithm. Briefly describe two ways to improve the time complexity of the algorithm.</p> <p>d) Given a grid of $(m \times n)$ dimension containing cells filled with reward (positive or negative), the reward of an area will be sum of rewards of all the cells within the area. If a problem is defined as to find the maximum possible reward from a minimum square:</p> <ul style="list-style-type: none"> i. Can you use dynamic programming for the problem defined above? If yes, what will be the optimal substructure for the problem? ii. If dynamic programming cannot be used or the benefit of using the algorithm is hindered for some reason, explain why it might be the case. | <p>1×5
(CO2)
(PO1)</p> <p>3
(CO2)
(PO1)</p> <p>3
(CO2)
(PO2)</p> <p>5
(CO2)
(PO2)</p> |

3. a) What is reduction? Why is it important in proving NP-completeness of a problem. 4
(CO2)
(PO2)
- b) Prove that every Formula SAT is reducible to 3-CNF SAT. 4
(CO3)
(PO2)
- c) Prove that circuit SAT is NP-complete. 4
(CO3)
(PO2)
- d) Prove that 2-CNF SAT is in P. 4
(CO3)
(PO2)

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SUMMER SEMESTER, 2023-2024
FULL MARKS: 75

CSE 4807: IT Organization and Management

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Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

-
1. It may surprise you that two college students working from their dorm room have built a communication platform that now serves over \$150 million users globally. Originally designed to help gamers communicate more effectively, their platform, *Discord*, has grown into a major player in online communities and professional collaboration. Jason Citron, co-founder and CEO, has led the company to become a multi-billion-dollar platform used by creators, businesses, and educators. *Discord* offers a free model with premium features, enabling it to serve a wide audience while generating significant revenue. The business primarily operates as a platform provider, allowing users and organizations to build their own communities. The business model focuses on dealing with other businesses. Given the scenario, answer the following questions:
- a) As the CEO of *Discord*, what are the skill sets that Jason Citron requires? Justify your answer. 4
(CO1)
(PO1)
- b) According to Henry Mintzberg, a prominent management researcher at McGill University, a manager has to perform certain roles in an organization. Considering *Discord* as an organization which conducts business with other business organizations, what should be the roles of Jason Citron in orchestrating such a successful organization as *Discord*? 10
(CO1)
(PO1)
- c) How many different types of international organizations are there and what type of organization is *Discord* in your opinion? Explain your answer. 3 + 3
(CO1)
(PO1)
- d) Environmental uncertainty can be defined as the degree of change and complexity in an organizations environment. Explain it by environmental uncertainty matrix. 5
(CO1)
(PO1)
-
2. A small-organic skincare company located in New Zealand has built a strong brand reputation for its handmade, eco-friendly products. It has a loyal customer base domestically and excellent online reviews. The company wants to expand its market to Southeast Asia but has limited financial resources and no international retail experience. The owner wants to retain control over product quality, brand image, and marketing messages, while also learning from partners who understand the target markets better.
- a) Mention and justify which type of business operation (importing, exporting, licensing, franchising, strategic alliance) would be the most appropriate for the above mentioned skincare company. 8
(CO2)
(PO1)
- b) Hofstede discovered some dimensions for helping manager to understand different national culture. Analyze these dimensions in the context of different countries. 10
(CO1)
(PO1)

- c) Stakeholders can significantly shape an organization's success or failure. Who are the primary stakeholders of an organization, and in what ways do their expectations drive organizational behavior and decision-making? 7
(CO1)
(PO1)
3. a) What are the various tools and techniques managers use to maximize the benefits of planning? 5
(CO1)
(PO1)
- b) Define the terms project, program, portfolio, and operations management and explain the relationship between programs, projects, shared resources, and stakeholders with a sample portfolio structure. 4 + 4
(CO1)
(PO1)
- c) PESTEL is an acronym of six factors. What are those factors and why is PESTEL analysis done in strategic management? 6
(CO2)
(PO1)
- d) For COCOMO model, explain the graph for effort versus product size for three types of project. Consider a new project with an estimated size of 450 KLOC that has to be developed. Project manager has a choice of hiring from two pools of developers: Developers with very high application experience with very little experience in the programming language or Developers with very low application experience but a lot of experience with the programming language. What is the impact of hiring all developers from one or the other pool? 2 + 4
(CO3)
(PO2)

Table 1: Value of Cost Drivers for Question 3.d

Personal Attributes	Very low	low	Nominal	High	Very High
Analyst Capability	1.46	1.19	1.00	0.86	0.71
Application Experience	1.29	1.13	1.00	0.91	0.82
Software Engineer Capability	1.42	1.17	1.00	0.86	0.70
Virtual Machine Experience	1.21	1.10	1.00	0.90	
Programming language Experience	1.14	1.07	1.00	0.95	

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CSE 4835: Pattern Recognition

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Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Justify your agreement/disagreement with the following statements: 3 × 4
(CO2)
(PO2)
 - i. Training on higher-level image features (e.g., color histograms or HOG features) instead of raw pixels improves image classification accuracy across all types of models.
 - ii. If the datapoints are linearly separable, then training a linear classifier with a Softmax loss and no regularization will achieve zero training loss at convergence.
 - iii. Given a constant gradient at each time step, the update to the weights will decrease over time when using the AdaGrad optimizer.
 - iv. Consider the Loss function given in Equation 1, where $R(w)$ is a $L2$ regularizer and L represents standard training loss. If we set $\lambda = -1$, then for every w , there exists another set of weights w' with lower loss, but an accuracy less than or equal to that of w .
$$L(w) = \frac{1}{N} \sum_{i=1}^N L_i(f(x_i, w), y_i) + \lambda R(w) \quad (1)$$
- b) Draw a computational graph for the function given in Equation 2. Calculate the values of each $\frac{df}{dw_i}$ and $\frac{df}{db_i}$ considering $w_0 = 1, w_1 = 3, w_2 = 4, x_0 = 3, x_1 = 2, b_0 = 8, b_1 = 1$. 8
(CO1)
(PO1)

$$f(x_0, x_1, w_0, w_1, w_2, b_0, b_1) = \max(w_2 \times (w_0 \times x_0 + w_1 \times x_1 + b_0) + b_1, 0) \quad (2)$$
2. a) Explain the significance of using an activation function using the concept of space warping. 7
(CO1)
(PO1)
- b) 'A Neural Network with one hidden layer can approximate any function $f : R^N \rightarrow R^M$ with arbitrary precision' — Justify the statement. 10
(CO2)
(PO2)
- c) Discuss the properties of Adam optimizer that contribute to its widespread adoption as a default optimization algorithm across diverse downstream tasks. 8
(CO1)
(PO1)
3. a) i. ResNet-50 represents an enhanced variant of ResNet-34, achieving a 1.45% reduction in the Top-5 error rate on the ImageNet dataset, while incurring only an additional 0.2 GFLOPs of computational overhead. Discuss the architectural design choices that contributed to this performance improvement. 5
(CO2)
(PO2)
- ii. The VGG16 and VGG19 both have five convolutional stages. The VGG19 architecture has additional convolution layers in the later stages. Assume these additional layers are introduced in the earlier stages instead of later. Criticize whether this design choice will improve computational efficiency and/or classification performance. 5
(CO2)
(PO2)

- b) Consider a 3D convolutional network with its layers specified in Table 1. For each layer, fill the blank fields in the table with the size of the activation volumes, and the number of weight and bias parameters. The INPUT layer is provided; this network takes in 8 frames of 64-by-64 pixel RGB images. Write your answer as a multiplication (e.g. $128 \times 128 \times 128$). The layer descriptions are as follows:

- CONV3-N-p1-s2 is a convolutional layer with N $3 \times 3 \times 3$ kernels, padding 1 and stride 2 (in height, width, and temporal dimension) with ReLU activation.
- POOL2 denotes a $2 \times 2 \times 2$ max-pooling layer, with stride 2 and no padding.
- FC-N denotes a fully-connected layer with N neurons, with ReLU activation.

Table 1: Layer description for Question 3.b

Layer	Activation Volume	Learnable Weight Parameters	Biases
INPUT	$3 \times 8 \times 64 \times 64$	N/A	N/A
CONV5-16-p2-s1			
POOL2			
CONV3-16-p1-s1			
POOL2			
CONV3-32-p1-s2			
FC-256			
FC-10			

- c) In a 3D convolutional network, we can define spatiotemporal batch normalization similarly to operate on a batch of 4D feature maps of shape $N \times C \times T \times H \times W$, normalizing over the N, T, H , and W dimensions. If we now add one spatiotemporal Batch Normalization (BN) after each convolution layer, how many additional parameters will be introduced in Table 2?

Table 2: Number of BN parameters for Question 3.c

Layer	BN Parameters
CONV5-16-p2-s1	
CONV3-16-p1-s1	
CONV3-32-p1-s2	

- d) Figure 1 shows an image of size 16×16 (partially shown). Design a 5×5 convolutional filter that is maximally activated when sliding over the '3' in Figure 1. Consider the black pixels as 1s, and the white pixels are -1. For simplicity, use only -1s and 1s in your designed filter.

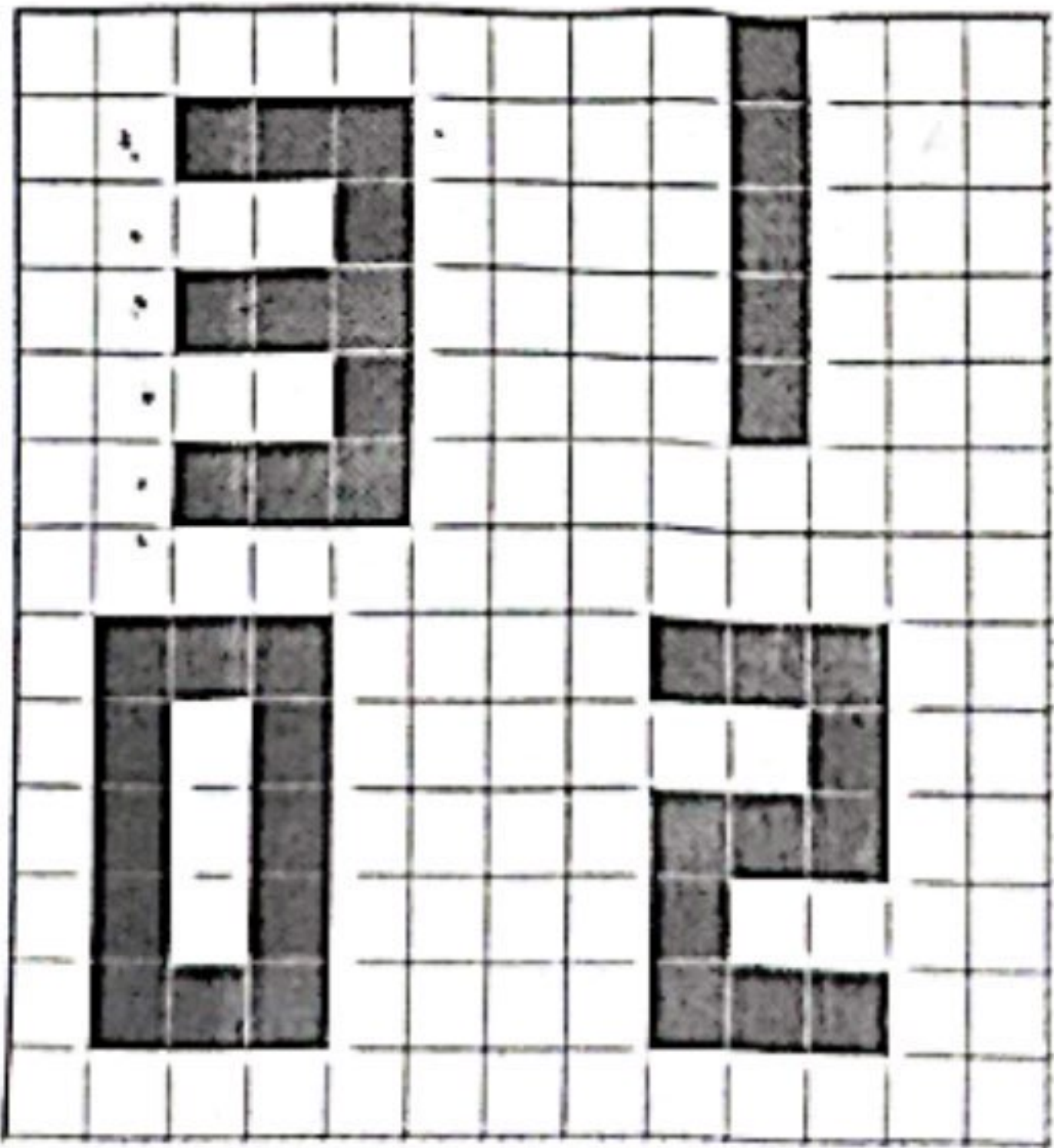


Figure 1: Sample image for Question 3.d